

Horn telescope design and construction

A classroom radio telescope

Adapted from Daniel Bonnett, DSPIRA 2018

Chandler Academy, Kanawha County, West Virginia

Telescope design

Radio astronomy

The flared horn collects radio waves.

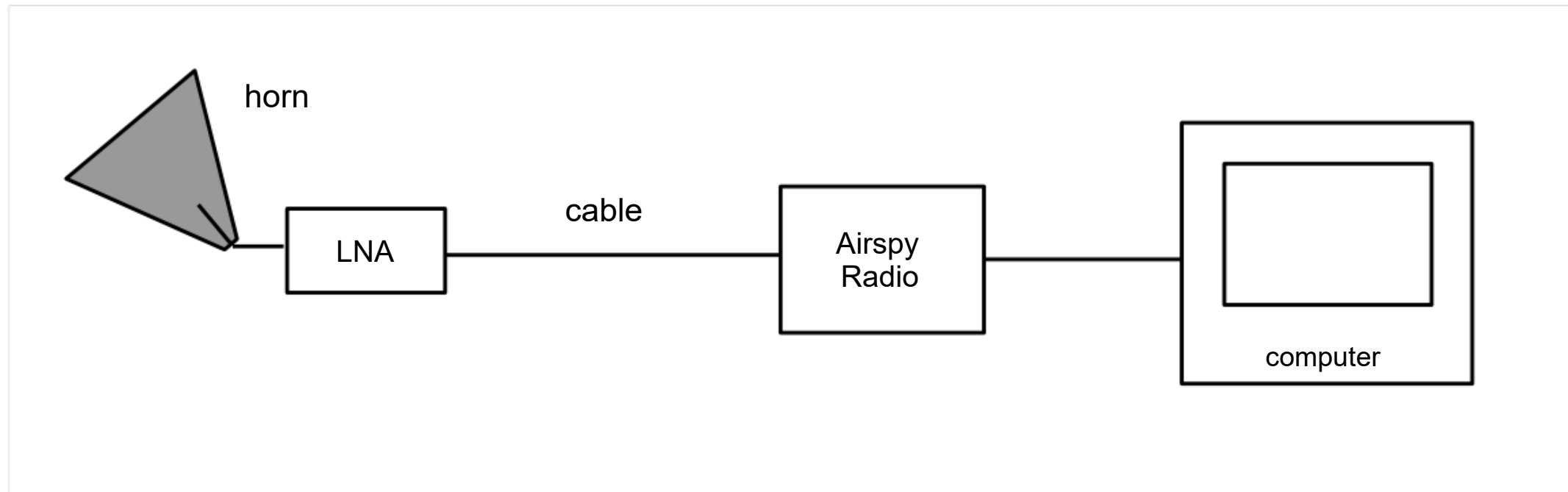
Its shape controls the telescope's response across the sky.

Classroom construction

Insulation board and a metal can make the telescope's receiver.

A wooden stand supports the telescope.

The receiver signal path



The low-noise amplifier (LNA) boosts the signal before the radio sends it to the computer.

Construction materials

Horn and receiver

Foil-faced insulation board

New, empty F-style gallon can

Copper probe and SMA connector

Foil tape and duct tape

Stand and cradle

Lumber and construction screws

Bolts, washers, and wing nuts

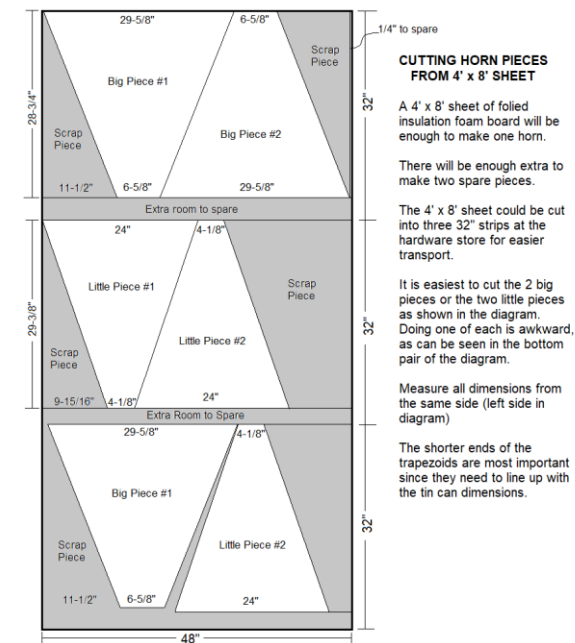
Use the linked guides for quantities and dimensions.

Horn panel cutting plan

Cut two large panels and two small panels.

Use the 2021 cutting guide for measurements.

One 4 × 8-foot sheet makes one horn.



2021 cutting plan. Open the full-size guide before cutting.

Joining the horn panels

Alternate large and small panels with the foil faces up.

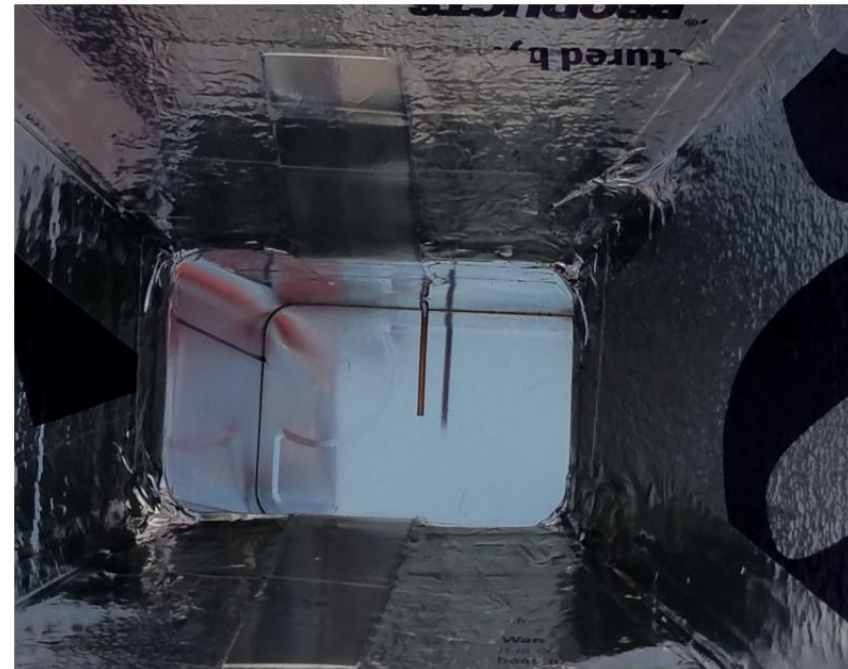
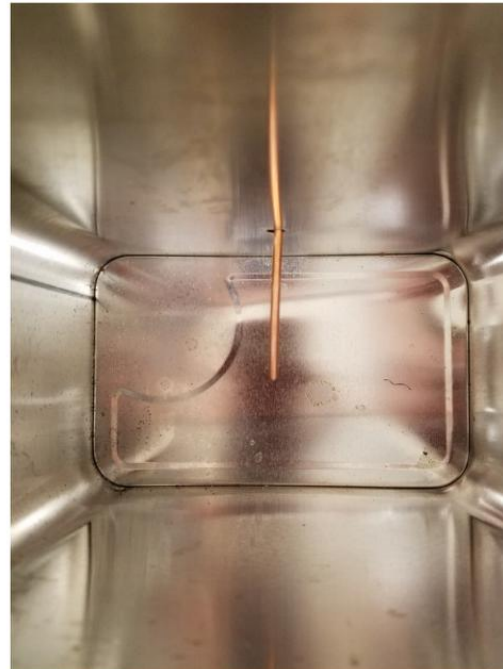
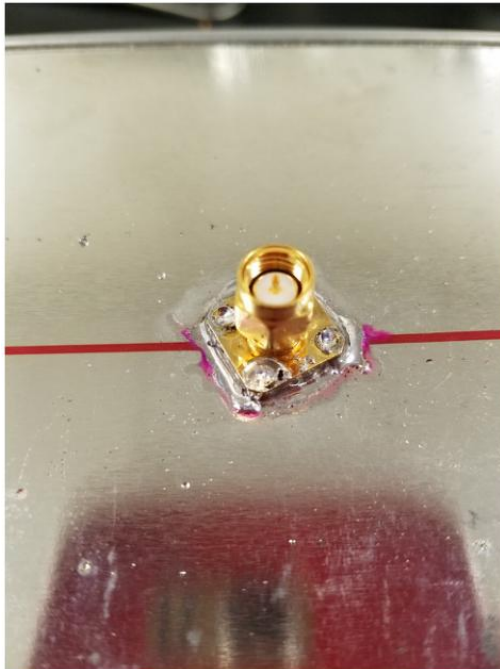
Tape the adjoining edges, then fold the horn.

Seal the interior seams with foil tape.



Panels before folding. Photo: DSPIRA 2018.

The can receiver



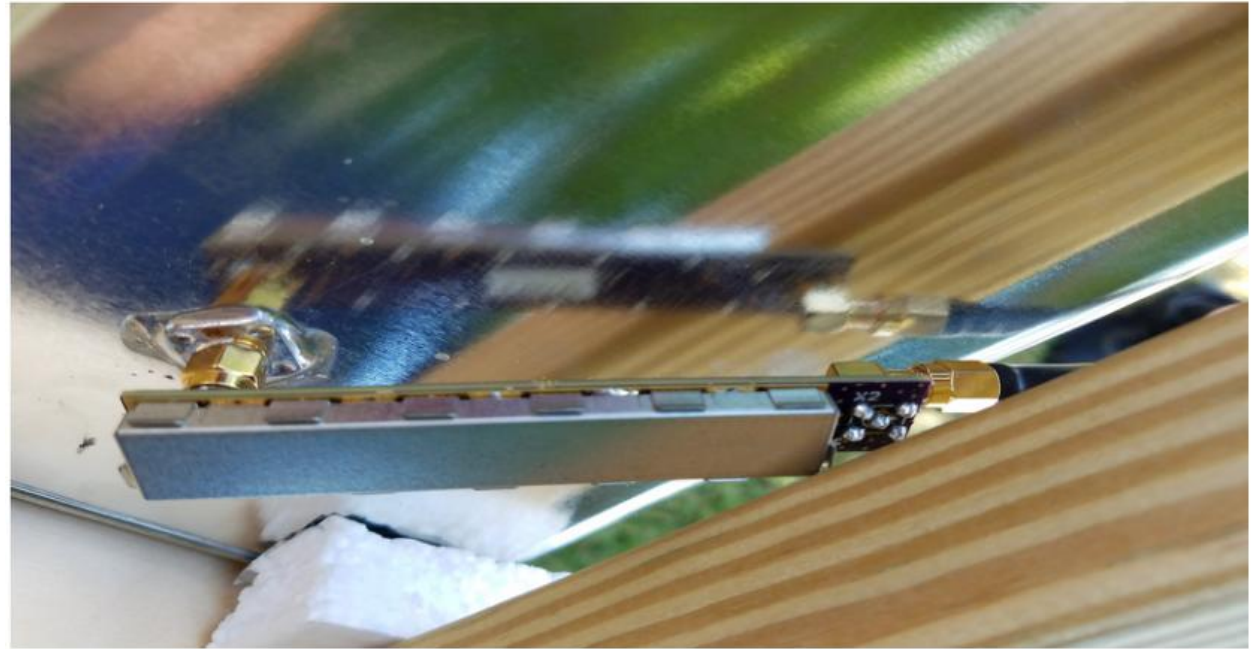
Use a new, empty can. The probe extends inside; the SMA connector stays outside. Photos: DSPIRA 2018.

The low-noise amplifier

The LNA mounts directly on the can's SMA connector.

The cable connects the LNA to the radio.

Follow the receiver guide for power and connection settings.



LNA at the receiver. Photo: DSPIRA 2018.

The telescope stand



Matching base notches and diagonal braces support the stand. Photos: DSPIRA 2018.

The horn cradle

The cradle holds the horn and adjusts its elevation.

Bolts, washers, and wing nuts connect it to the stand.

Check stability whenever you change the angle.



Cradle and stand. Photo: DSPIRA 2018.

Construction guides

[Horn and can assembly](#)

[Stand assembly](#)

[Cradle assembly](#)

[Mini horn construction](#)